

CHAPTER 5

STRUTS





TOPICS TO BE COVERED

5.1 Introduction

5.2 Euler's Theory

5.3 Equivalent strut length

5.4 Comparison of Euler's theory with experimental results

5.5 Euler's validity limit

5.6 Rankine or Rankine-Gordon formula

5.7 Struts with Initial Curvature

5.8 Struts with Eccentric Load

5.9 Laterally Loaded Struts

5.1 INTRODUCTION

- Structural members which carry compressive loads may be divided into two broad categories depending on their relative lengths and cross-sectional dimensions.
- Short members are generally termed **Columns** and these usually fail by *crushing* when the stress of the material in compression is exceeded.
- Long, slender columns or **Struts**, however, fail by buckling some time before the yield stress in compression is reached.

- The Buckling occurs owing to one or more of the following reasons:

- A) The strut may not be perfectly straight initially;
- B) The load may not be applied exactly along the axis of the strut;
- C) One part of the material may yield in compression more readily than others owing to some lack of uniformity in the material properties throughout the strut.

- At values of load below the buckling load a strut will be in stable equilibrium where the displacement caused by any lateral disturbance will be totally recovered when the disturbance is removed.
- At the buckling load the strut is said to be in a state of **neutral equilibrium**, and theoretically it should then be possible to gently deflect the strut in to a **simple sine wave** provided that the amplitude of the wave is kept small.

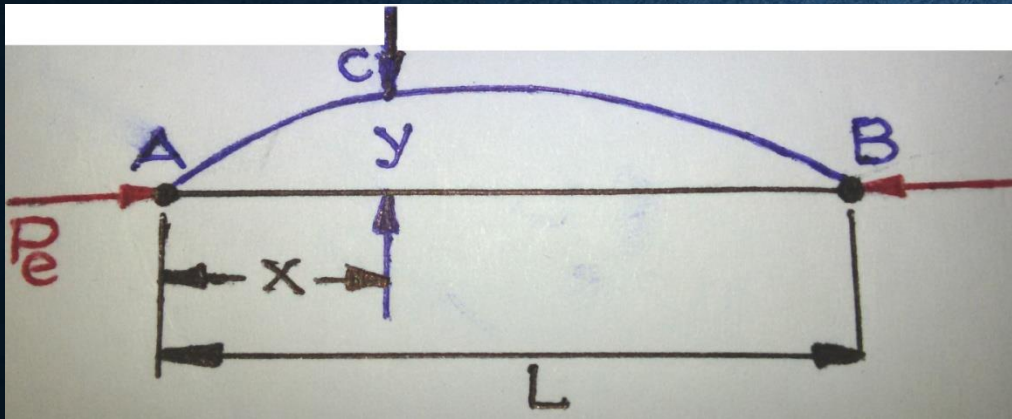
Example: Thin Metal strips

- Theoretically, it is possible for struts to achieve a condition of **unstable equilibrium** with loads exceeding the buckling load, any slight lateral disturbance then then causing failure by Buckling; this condition is never achieved in practice under static load conditions.
- **Buckling** occurs immediately at the point where the buckling load is reached owing to the reasons stated earlier.

5.2 EULER'S THEORY

A. Strut with pinned ends

Consider the axially loaded strut shown in **Figure 5.1** subjected to the crippling load P_e producing a deflection y at a distance x from one end. Assume that the ends are either pin-jointed or rounded so that there is no moment at either end.



$$\text{B.M at C} = EI \frac{d^2y}{dx^2} = -P_e y$$

$$EI \frac{d^2y}{dx^2} + P_e y = 0$$

$$\frac{d^2y}{dx^2} + \frac{P_e}{EI} y = 0$$

Fig.5.1 Strut with axial load and pinned ends

- Let $D=d/dx$,

$$(D^2 + n^2)y = 0, \text{ where } n^2 = P_e / EI$$

- This is a second-order differential equation which has a solution of the form

$$y = A \cos nx + B \sin nx$$

$$\text{i.e. } y = A \cos \sqrt{\left(\frac{P_e}{EI}\right)}x + B \sin \sqrt{\left(\frac{P_e}{EI}\right)}x$$

Now at $x=0, y=0$

$$A=0$$

And at $x=L, y=0$

$$B \sin L \sqrt{(P_e / EI)} = 0$$

Either $B=0$ or $\sin L \sqrt{(P_e / EI)} = 0$

- If $B=0$ then $y=0$ and the strut has not yet **Buckled**. Thus the solution required is:

$$\sin L\sqrt{\left(\frac{P_e}{EI}\right)} = 0 \quad \therefore L\sqrt{\left(\frac{P_e}{EI}\right)} = \pi$$

$$P_e = \frac{\pi^2 EI}{L^2} \dots\dots\dots \text{EQ.1}$$

- It should be noted that other solutions(infinite number of solutions) exist for the equation

$$\sin L\sqrt{\left(\frac{P_e}{EI}\right)} = 0 \quad \text{i.e } \sin nL = 0$$

$$nL = \pi, 2\pi, 3\pi, 5\pi, \text{etc}$$

- The above selected value in equation 1 is the so-called *Fundamental* mode value and is the lowest critical load producing the single-bow buckling condition.
- The solution $nL=2\pi$ produces in two half waves, 3π in three half-waves etc as shown in Figure 5.2.
- If the load is applied sufficiently to the strut, it is possible to pass through the *fundamental mode* and to achieve at least one of the other modes which are theoretically possible.

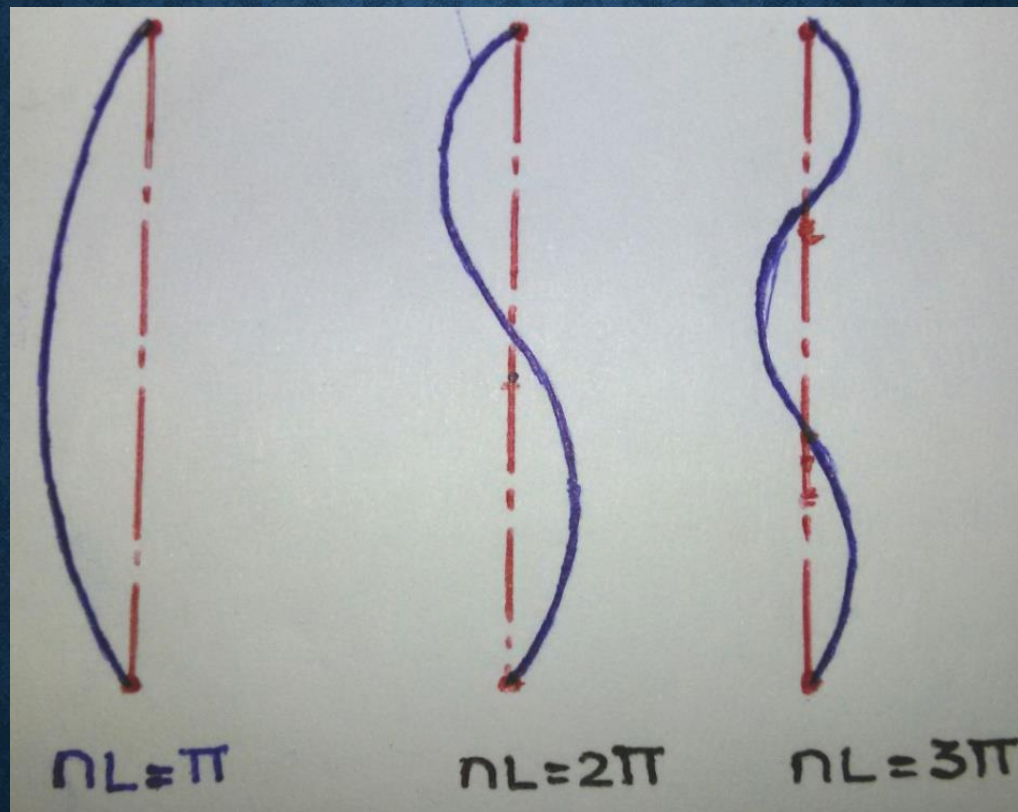


Fig 5.2: Strut Failure Modes

B. One end fixed, the other free

Consider now the strut of *Figure 5.3* with the origin at the fixed end.

$$\text{B.M at C} = EI \frac{d^2y}{dx^2} = +P(a - y)$$

$$\frac{d^2y}{dx^2} + \frac{Py}{EI} = \frac{Pa}{EI}$$

$$(D^2 + n^2)y = n^2a \dots \dots \dots \text{Eq.2}$$

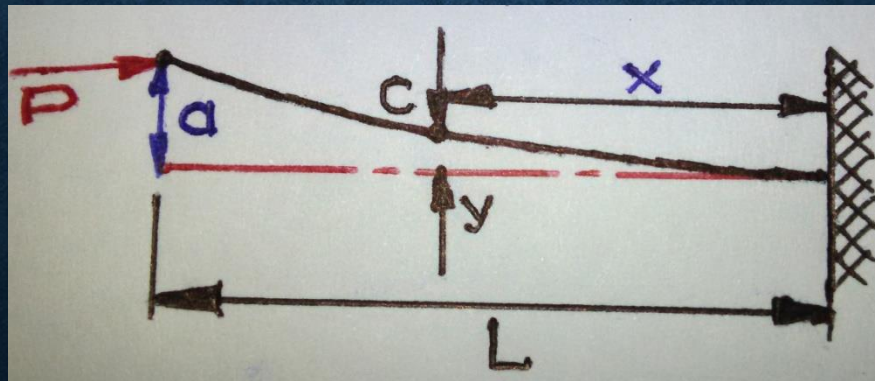


Fig. 5.3: Fixed-free strut

N.B: It is always convenient to arrange the diagram and origin such that the differential equation is achieved in the above form since the solution will then always be of the form.

$$y=A \cos nx+B \sin nx + (\textit{Particular solution})$$

The particular solution is a particular value of y which satisfies *equation 2*, and in this case can be shown to be $y=a$.

$$y=A \cos nx+B \sin nx+a$$

Now when $x=0$, $y=0$

$$A=-a$$

When $x=0$, $dy/dx=0$

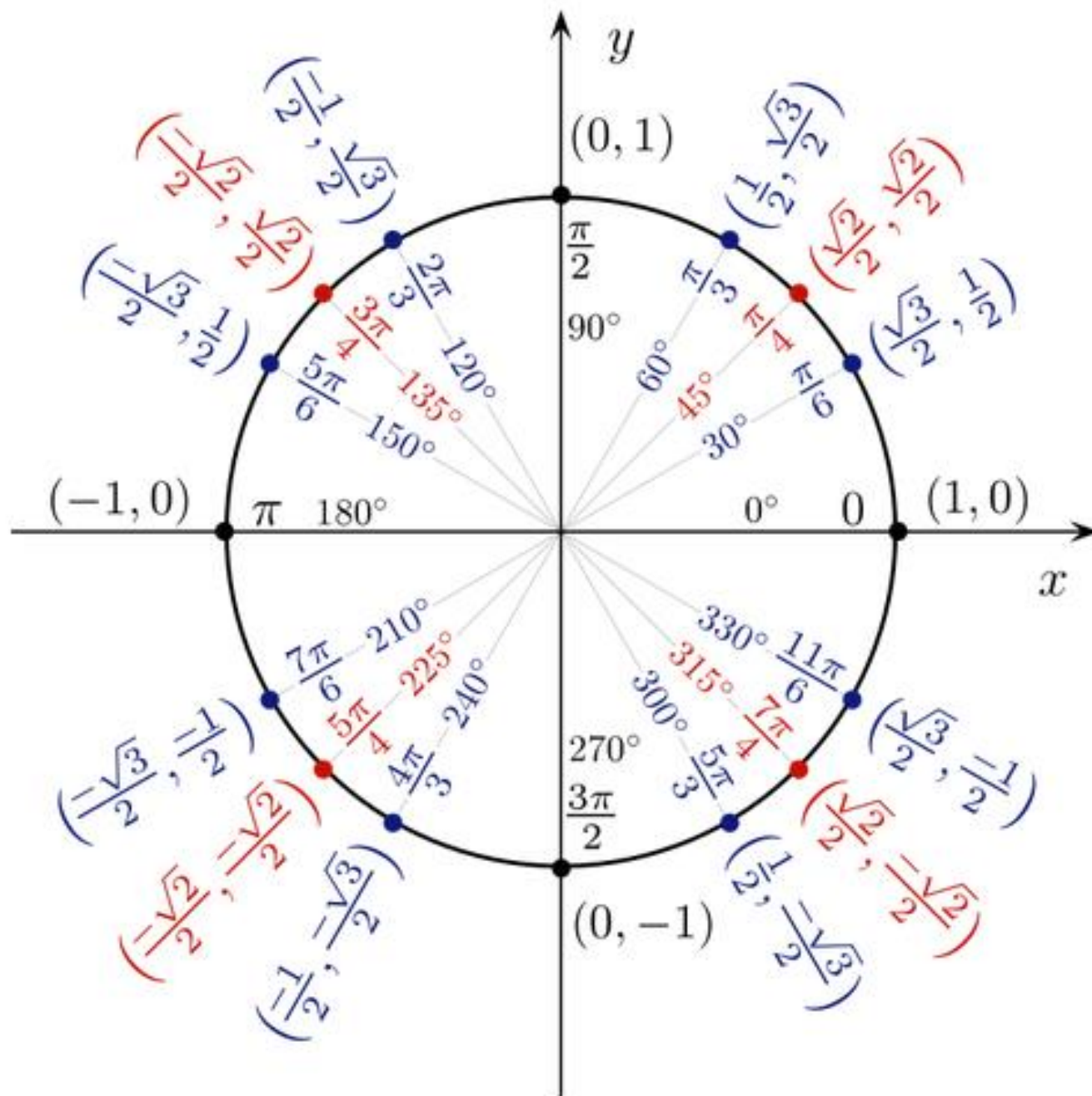
$$B=0$$

$$y= -a \cos nx+a$$

But when $x=L$, $y=a$

$$a= -a \cos nL+a$$

$$0= \cos nL$$



- The fundamental mode of buckling in this case therefore is given when $nL=(1/2)\pi$.

$$L\sqrt{\left(\frac{P}{EI}\right)} = \frac{\pi}{2}$$
$$P_e = \frac{\pi^2 EI}{4L^2} \dots\dots\dots \text{Eq. 3}$$

C. Fixed Ends

Consider the strut of Figure 5.4 with the origin at the centre.

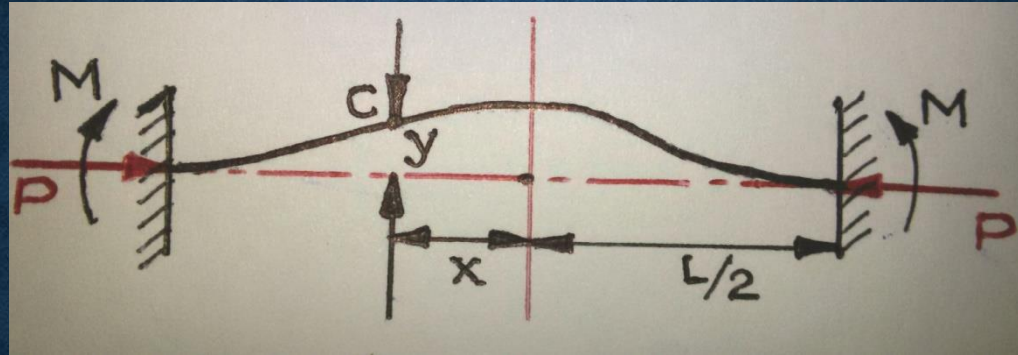


Fig 5.4: Strut with fixed ends

In this case the B.M at C is given by:

$$EI \frac{d^2 y}{dx^2} = M - Py$$

$$\frac{d^2 y}{dx^2} + \frac{P}{EI} y = \frac{M}{EI}$$

$$(D^2 + n^2)y = M/EI$$

- Here the particular solution is:

$$y = \frac{M}{n^2 EI} = \frac{M}{P}$$

$$y = A \cos nx + B \sin nx + M / P$$

- Now when $x=0$, $dy/dx=0$ Therefore, $B=0$ and when $x=1/2L$, $y=0$

$$\therefore A = -\frac{M}{P} \sec \frac{nL}{2}$$

$$y = -\frac{M}{P} \sec \frac{nL}{2} \cos nx + \frac{M}{P}$$

- But when $x=1/2L$, dy/dx is also zero,

$$0 = \frac{nM}{P} \sec \frac{nL}{2} \sin \frac{nL}{2}$$

$$0 = \frac{nM}{P} \tan \frac{nL}{2}$$

- The fundamental buckling mode is then given when $nL/2=\pi$.

$$\frac{L}{2} \sqrt{\left(\frac{P}{EI}\right)} = \pi$$

$$P_e = \frac{4\pi^2 EI}{L^2} \dots\dots\dots \text{Eq.4}$$

D. One end fixed, the other Pinned

In order to maintain the pin-joint on the horizontal axis of the unloaded strut, it is necessary in this case to introduce a vertical load F at the Pin (**Figure 5.5**). The moment of F about the built-in end then balances the fixing moment.

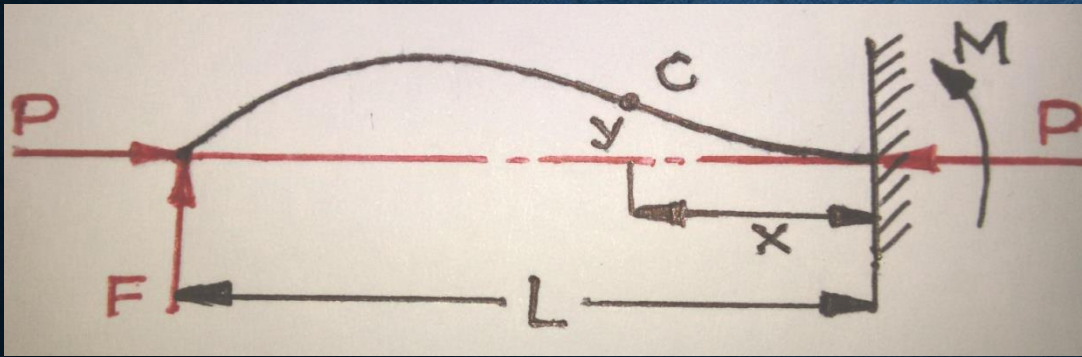


Fig 5.5: Strut with one end pinned,
the other fixed

$$EI \frac{d^2 y}{dx^2} = -Py + F(L - x)$$

$$\frac{d^2 y}{dx^2} + \frac{P}{EI} y = \frac{F}{EI} (L - x)$$

$$(D^2 + n^2)y = \frac{F}{EI} (L - x)$$

- The particular solution is:

$$y = \frac{F}{n^2 EI} (L - x) = \frac{F}{P} (L - x)$$

- The full solution is therefore

$$y = A \cos nx + B \sin nx + \frac{F}{P} (L - x)$$

- When $x=0$, $y=0$, Therefore, $A = -FL/P$
- When $x=0$, $dy/dx=0$, $B = F/np$

$$\begin{aligned} y &= -\frac{FL}{P} \cos nx + \frac{F}{nP} \sin nx + \frac{F}{P} (L - x) \\ &= \frac{F}{nP} [-nL \cos nx + \sin nx + n(L - x)] \end{aligned}$$

- But when $x=L$, $y=0$

$$nL \cos nL = \sin nL$$

$$\tan nL = nL$$

- The lowest value of nL (neglecting zero) which satisfies this condition and which therefore produces the fundamental buckling condition is **$nL=4.5$ radians.**

$$L\sqrt{\frac{P}{EI}} = 4.5$$

$$P_e = \frac{20.25EI}{L^2} \dots\dots\dots \text{Eq.5}$$

or approximately

$$P_e = \frac{2\pi^2 EI}{L^2} \dots\dots\dots \text{Eq.6}$$

5.3 EQUIVALENT STRUT LENGTH

- Having derived the result for the buckling load of a strut, the Euler loads for other end conditions may all be written in the same form,

i.e

$$P_e = \frac{\pi^2 EI}{l^2} \dots\dots\dots \text{Eq.7}$$

Where l is the *equivalent length* of the strut and can be related to the actual length of the strut depending on the end conditions. The equivalent length is the length of a single bow (half sine wave) in each of the strut deflection curves shown in the figure below.

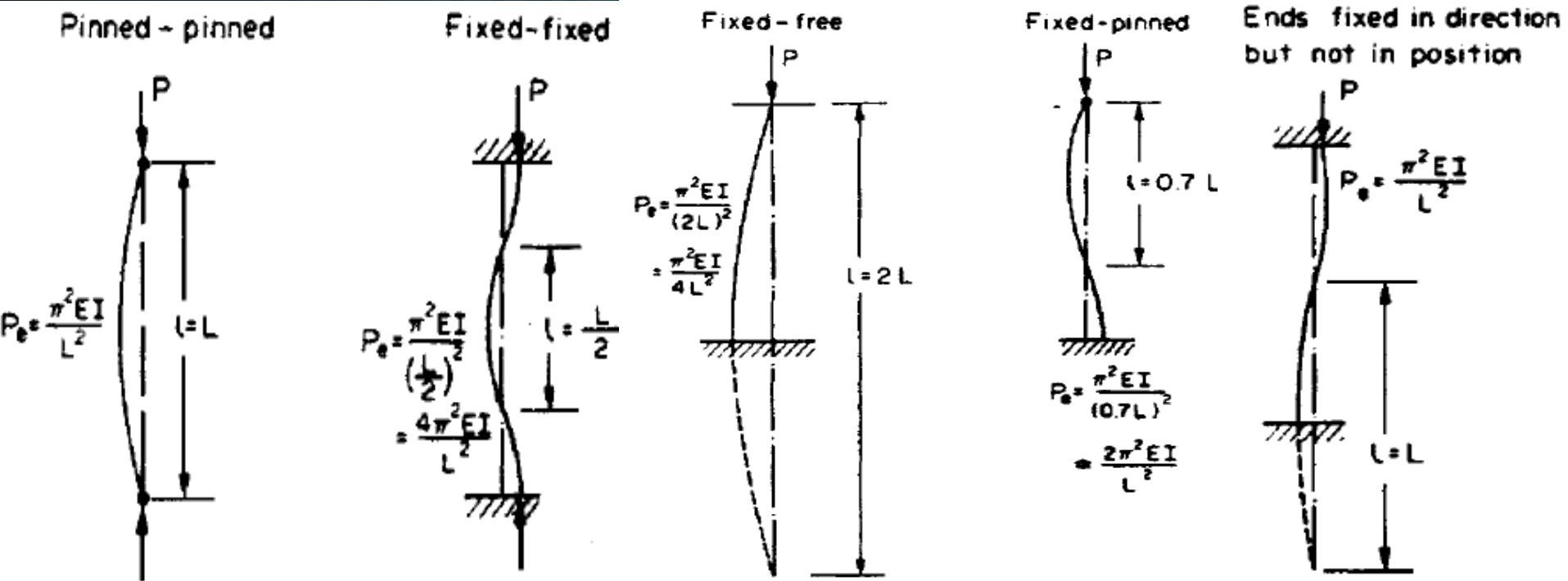


Figure 5.6: “Equivalent length” of struts with different end conditions. In each case l is the length of a single bow.

5.4 COMPARISON OF EULER'S THEORY WITH EXPERIMENTAL RESULTS

- Between $L/k=40$ and $L/k=100$, the Euler results are not close to the experimental values, suggesting a critical load which is in excess of what is actually required for failure - a very unsafe situation!
- Other formulae have therefore been derived in an attempt to obtain closer agreement between the actual failing load and the predicted value in this range of slenderness ratio.

a) Straight-line formula

$$P = \sigma_y A [1 - n(L/k)] \dots \dots \dots \text{Eq.8)}$$

The value of n depending on the material used and the end condition.

b) Johnson Parabolic formula

$$P = \sigma_y A [1 - b(L/k)^2] \dots \dots \dots \text{Eq.9)}$$

The value of b depending also on the end condition.

Neither of the above formulae proved to be very successful, and they were replaced by:

C. Rankine-Gordon formula

$$\frac{1}{P_R} = \frac{1}{P_e} + \frac{1}{P_C} \dots \dots \dots \text{Eq.10}$$

Where P_e is the Euler buckling load and P_c is the crushing (compressive yield) load $= \sigma_y A$.

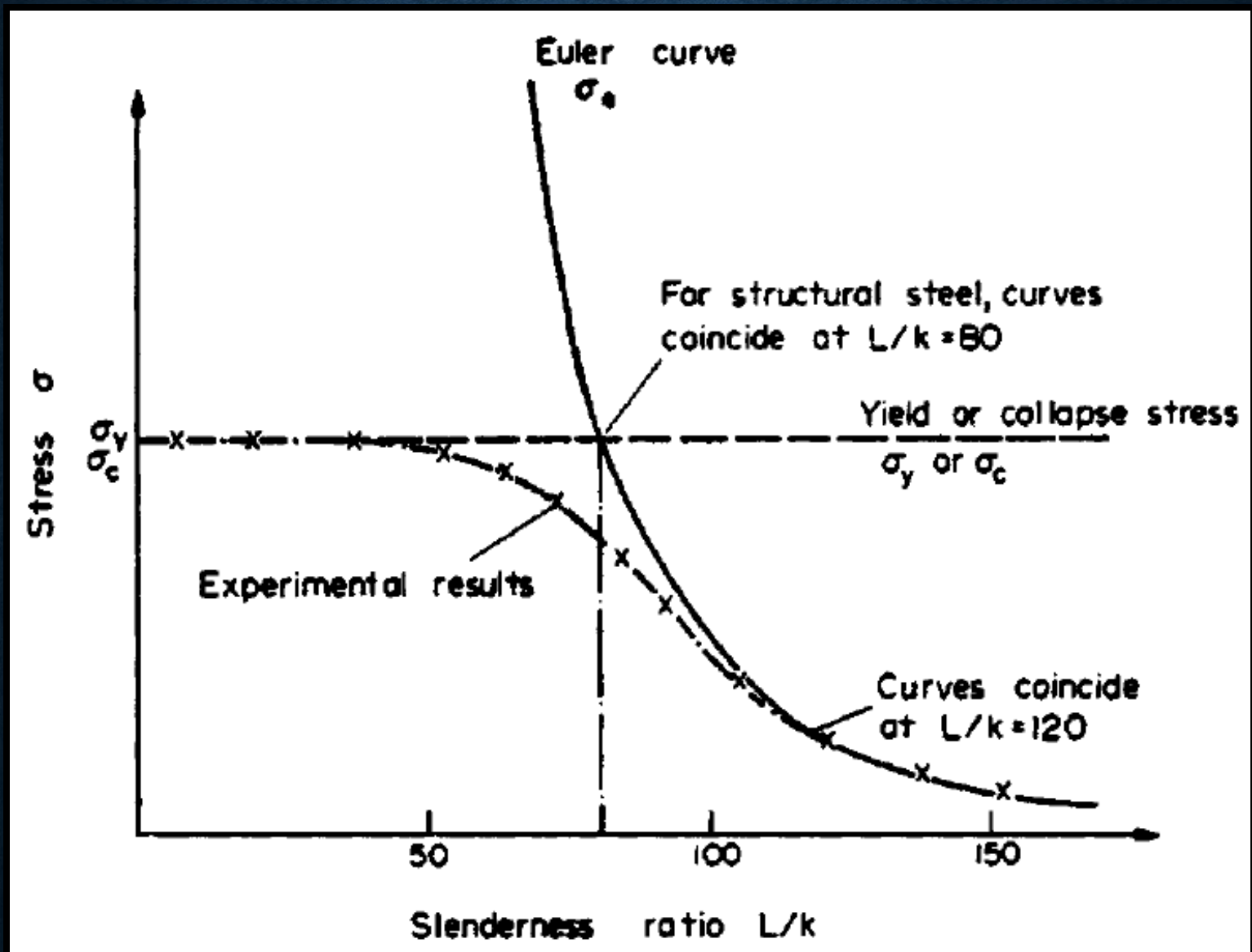


Fig 5.7: Comparison of experimental results with Euler curve

5.5 EULER “VALIDITY LIMIT”

- From the graph of *Fig. 5.7* and the comments above, it is evident that the Euler theory is unsafe for small L/k ratios. It is useful, therefore, to determine the limiting value of L/k below which the Euler theory should not be applied; this is termed the *validity limit*.
- The validity limit is the point where the Euler stress σ_e equals the yield or crushing stress σ_y , i.e the point where the strut load
- Now the Euler load can be written in the form

$$P = \sigma_y A$$

- Where C is a constant depending on the end condition of the strut.

$$P_e = C \frac{\pi^2 EI}{L^2} = C \frac{\pi^2 E A k^2}{L^2}$$

- Therefore in the limiting condition

$$\sigma_y A = C \frac{\pi^2 E A k^2}{L^2}$$
$$\frac{L}{k} = \sqrt{\left(\frac{C \pi^2 E}{\sigma_y} \right)}$$

- The value of this expression will vary with the type of end condition; as an example, low carbon steel struts with pinned ends give $L/k \approx 80$.

5.6 Rankine or Rankine-Gordon formula

- The Rankine formula is a combination of the Euler and crushing loads for a strut

$$\frac{1}{P_R} = \frac{1}{P_e} + \frac{1}{P_C}$$

- For very short struts P_e is very large; $1/P_e$ can therefore be neglected and $P_R = P_C$. For very long struts P_e is very small and $1/P_e$ is very large so that $1/P_C$ can be neglected. Thus $P_R = P_e$.
- The Rankine formula is therefore valid for extreme values of L/k . It is also found to be fairly accurate for the intermediate values in the range under consideration.

- Thus, re-writing the formula in terms of stresses,

$$\frac{1}{\sigma A} = \frac{1}{\sigma_e A} + \frac{1}{\sigma_y A}$$

$$\text{i.e. } \frac{1}{\sigma} = \frac{1}{\sigma_e} + \frac{1}{\sigma_y} = \frac{\sigma_e + \sigma_y}{\sigma_e \sigma_y}$$

$$\sigma = \frac{\sigma_e \sigma_y}{\sigma_e + \sigma_y} = \frac{\sigma_y}{[1 + (\sigma_y / \sigma_e)]}$$

- For a strut with both ends pinned

$$\sigma_e = \frac{\pi^2 E}{(L/k)^2}$$

$$\sigma = \frac{\sigma_y}{1 + \frac{\sigma_y}{\pi^2 E} (L/k)^2}$$

$$\text{Rankine stress } \sigma_R = \frac{\sigma_y}{1 + a(L/k)^2} \dots \dots \dots \text{Eq.11}$$

- Where $a = \sigma_y / \pi^2 E$, theoretically, but having a value normally found by experiment for various materials.

- Therefore, The Rankine load is :

$$P_R = \frac{\sigma_y A}{1 + a(L/k)^2} \dots\dots\dots \text{Eq.12}$$

- Typical values of a for use in the Rankine formula are given in **Table 5.1** below. However, since the values of a are not exactly equal to the theoretical values, the Rankine loads for long struts will not be identical to those estimated by the Euler theory as suggested earlier.

Material	σ_y or σ_c (MN/m ²)	a	
		Pinned ends	Fixed ends
Low carbon steel	315	1/7500	1/30 000
Cast iron	540	1/1600	1/64 000
Timber	35	1/3000	1/12 000

N.B. a for pinned ends = $4 \times (a$ for fixed ends)

Table 5.1 Value of 'a' for different materials

5.7 STRUTS WITH INITIAL CURVATURE

Perry-Robertson Formula

- The **Perry-Robertson** proof is based on the assumption that any imperfections in the strut, through **faulty workmanship** or **material** or **eccentricity of loading**, can be allowed for by giving the strut an initial curvature. For ease of calculation this is assumed to be a cosine curve, although the actual shape assumed has very little effect on the result.
- Consider, therefore, the strut AB of figure 5.8, of length L and pin-jointed at the ends. The initial curvature y_0 at any distance x from the centre is then given by :

$$y_0 = C_0 \cos \frac{\pi x}{L}$$

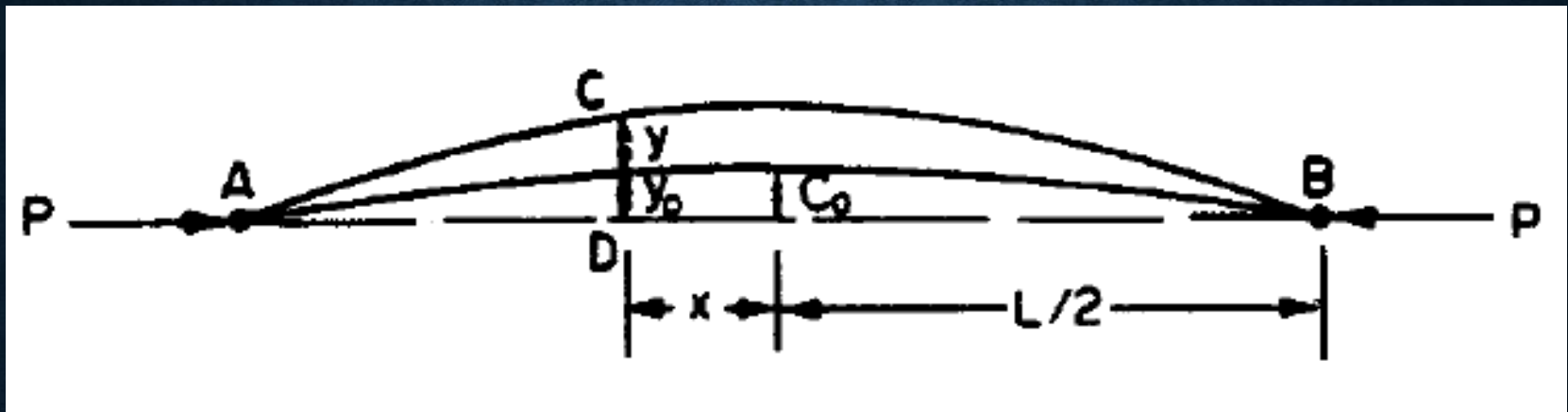


Figure 5.8: Strut with initial curvature

- If a load P is now applied at the ends, this deflection will be increased to $y+y_0$.

$$BM_C = EI \frac{d^2 y}{dx^2} = -P(y + C_0 \cos \frac{\pi x}{L})$$

$$\frac{d^2 y}{dx^2} + \frac{P}{EI} (y + C_0 \cos \frac{\pi x}{L}) = 0$$

The solution of which is

$$y = A \sin \sqrt{\left(\frac{P}{EI}\right)x} + B \cos \sqrt{\left(\frac{P}{EI}\right)x} + \left[\left(\frac{PC_0}{EI} \cos \frac{\pi x}{L}\right) / \left(\frac{\pi^2}{L^2} - \frac{P}{EI}\right)\right]$$

- Where A and B are the constants of integration.

Now when $x = \pm L/2$, $y = 0$

$$A = B = 0$$

$$y = [(\frac{PC_0}{EI} \cos \frac{\pi x}{L}) / (\frac{\pi^2}{L^2} - \frac{P}{EI})] = [(PC_0 \cos \frac{\pi x}{L}) / (\frac{\pi^2 EI}{L^2} - P)]$$

- Therefore dividing through, top and bottom, by A,

$$y = [(\frac{P}{A} C_0 \cos \frac{\pi x}{L}) / (\frac{\pi^2 EI}{L^2 A} - \frac{P}{A})]$$

- But $P/A = \sigma$ and $(\pi^2 EI)/(L^2 A) = \sigma_e$ (the Euler stress for pin-ended struts)

$$y = \frac{\sigma}{(\sigma_e - \sigma)} C_0 \cos \frac{\pi x}{L}$$

- Therefore total deflection at any point is given by :

$$\begin{aligned}
 y + y_0 &= \left[\frac{\sigma}{(\sigma_e - \sigma)} \right] C_0 \cos \frac{\pi x}{L} + C_0 \cos \frac{\pi x}{L} \\
 &= \left[\frac{\sigma_e}{(\sigma_e - \sigma)} \right] C_0 \cos \frac{\pi x}{L} \dots\dots\dots \text{Eq.13}
 \end{aligned}$$

- Maximum deflection (when x = 0), Bending Moment (BM) and Maximum stress due to bending are:

$$\text{Deflection} = \left[\frac{\sigma_e}{(\sigma_e - \sigma)} \right] C_0 \dots\dots\dots \text{Eq.14}$$

$$\text{B.M} = P \left[\frac{\sigma_e}{(\sigma_e - \sigma)} \right] C_0 \dots\dots\dots \text{Eq.15}$$

$$\text{Stress} = \frac{My}{I} = \frac{P}{I} \left[\frac{\sigma_e}{(\sigma_e - \sigma)} \right] C_0 h$$

where h is the distance of the outside fibre from the N.A of the strut

- Therefore the maximum stress owing to combined bending and axial is given by:

$$\begin{aligned}
 \sigma_{\max} &= \frac{P}{I} \left[\frac{\sigma_e}{(\sigma_e - \sigma)} \right] C_0 h + \frac{P}{A} \dots\dots\dots \text{Eq.16} \\
 &= \frac{P}{A k^2} \left[\frac{\sigma_e}{(\sigma_e - \sigma)} \right] C_0 h + \frac{P}{A} \\
 &= \sigma \left[\frac{\eta \sigma_e}{(\sigma_e - \sigma)} + 1 \right] \text{ where } \eta = \frac{C_0 h}{k^2}
 \end{aligned}$$

- If $\sigma_{\max} = \sigma_y$, the compressive yield stress for the material of the strut, the above equation when solved for σ gives

$$\sigma = \frac{[\sigma_y + (\eta + 1)\sigma_e]}{2} - \sqrt{\left\{ \left[\frac{\sigma_y + (\eta + 1)\sigma_e}{2} \right]^2 - \sigma_y \sigma_e \right\}}$$

(Perry-Robertson formula.(Eq.17))

- Thus for an initial curvature with a central deflection C_0 ,

$$\text{Maximum Deflection} = \left[\frac{\sigma_e}{(\sigma_e - \sigma)} \right] C_0 = \left[\frac{P_e}{P_e - P} \right] C_0 \dots \dots \text{Eq.19}$$

$$\text{Maximum B.M} = P \left[\frac{\sigma_e}{(\sigma_e - \sigma)} \right] C_0 = \left[\frac{PP_e}{P_e - P} \right] C_0 \dots \dots \dots \text{Eq.20}$$

$$\sigma_{\max} = \frac{P}{A} \pm \left[\frac{P\sigma_e}{(\sigma_e - \sigma)} \right] \frac{C_0 h}{I} = \frac{P}{A} \pm \left[\frac{PP_e}{P_e - P} \right] \frac{hC_0}{I} \dots \dots \dots \text{Eq.21}$$

where h is the distance of the outside fibre from the N.A of the strut

5.8 STRUTS WITH ECCENTRIC LOAD

- For eccentric loading at the ends of a strut, **Ayrton and Perry** suggest that the Perry-Robertson formula can be modified by replacing **C_0** by **$(C_0 + 1.2e)$** where e is the eccentricity.

$$\eta' = \eta + 1.2 \frac{eh}{k^2} \dots \dots \dots \text{Eq.22}$$

- And η' replaces η in the original **Perry-Robertson Equation**.

A. Pinned Ends -- The Smith-south well Formula

- Consider a strut loaded as shown in Figure 5.9. In this case there is strictly no 'buckling' load as previously described since the strut will bend immediately when the load is applied, bending taking place about the other principal axis.

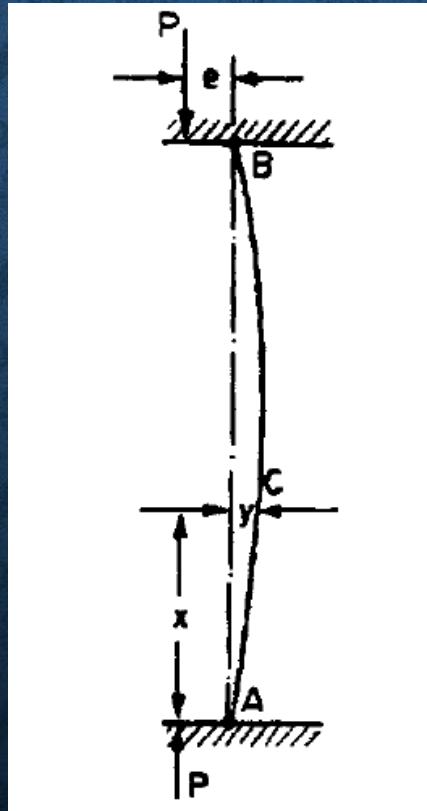


Figure 5.9 : Strut with eccentric load(Pinned Ends)

- Applying a similar procedure to that of the previous:

Maximum deflection,

$$\delta + e = e \frac{\sin^2 \frac{nL}{2}}{\cos \frac{nL}{2}} + e \cos \frac{nL}{2} = e \sec \frac{nL}{2} \dots\dots\dots \text{Eq.23}$$

$$\text{Maximum B.M} = P(\delta + e) = P e \sec \frac{nL}{2} \dots\dots\dots \text{Eq.24}$$

Maximum stress owing to bending

$$= \frac{M_y}{I} = P e \sec \frac{nL}{2} \times \frac{h}{I}$$

Where h is the distance from the N.A to the highest stressed fibre.

- Therefore the total maximum compressive stress to combined bending and thrust, assuming a **ductile material**, is given by [Smith-Southwell formula]

$$\begin{aligned}\sigma_{\max} &= \frac{P}{A} + (Pe \sec \frac{nL}{2}) \frac{h}{I} \\ &= \sigma [1 + \frac{eh}{k^2} \sec \frac{nL}{2}] \\ &= \sigma [1 + \frac{eh}{k^2} \sec \frac{L}{2} \sqrt{\frac{P}{EI}}] \dots \dots \dots \text{Eq.25}\end{aligned}$$

- For a **brittle material** (i.e weak in tension it is the maximum tensile stress which becomes the criterion for failure and the bending and direct stress components are opposite in sign.

$$\sigma_{\max} = \sigma [1 + \frac{eh}{k^2} \sec \frac{L}{2} \sqrt{(\frac{P}{Ek^2})}] \dots \dots \dots \text{Eq.26}$$

B. One End Fixed, The Other Free

- Consider the strut shown in Figure 2.10 below:

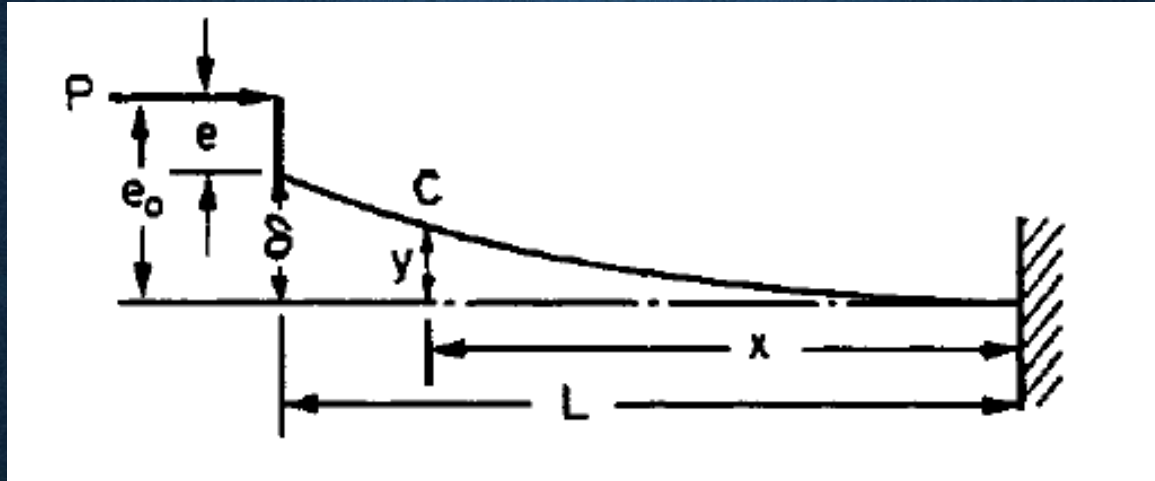


Figure 2.10: Strut with eccentric load (One fixed and free)

- In such case the Smith-Southwell formula can be written in the form

$$\sigma_{\max} = \sigma \left[1 + \frac{eh}{k^2} \sec \frac{L}{2} \sqrt{\left(\frac{\sigma}{Ek^2} \right)} \right] \dots \dots \dots \text{Eq.27}$$

5.9 LATERALLY LOADED STRUTS

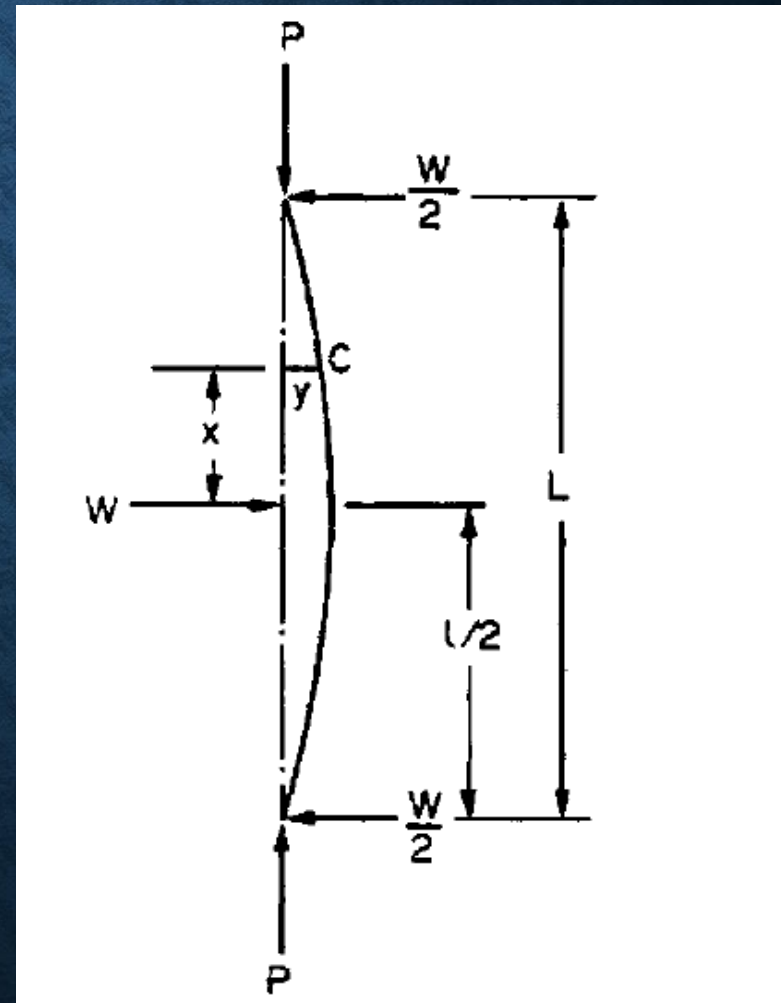
a. Central concentrated Load

With the origin at the centre of the strut as shown in Figure 5.11.

$$\text{B.M at C} = EI \frac{d^2y}{dx^2} = -Py - \frac{W}{2} \left(\frac{L}{2} - x \right)$$

$$\frac{d^2y}{dx^2} + n^2y = -\frac{W}{2EI} \left(\frac{L}{2} - x \right)$$

Figure 5.11



- The maximum deflection occurs where x is zero,

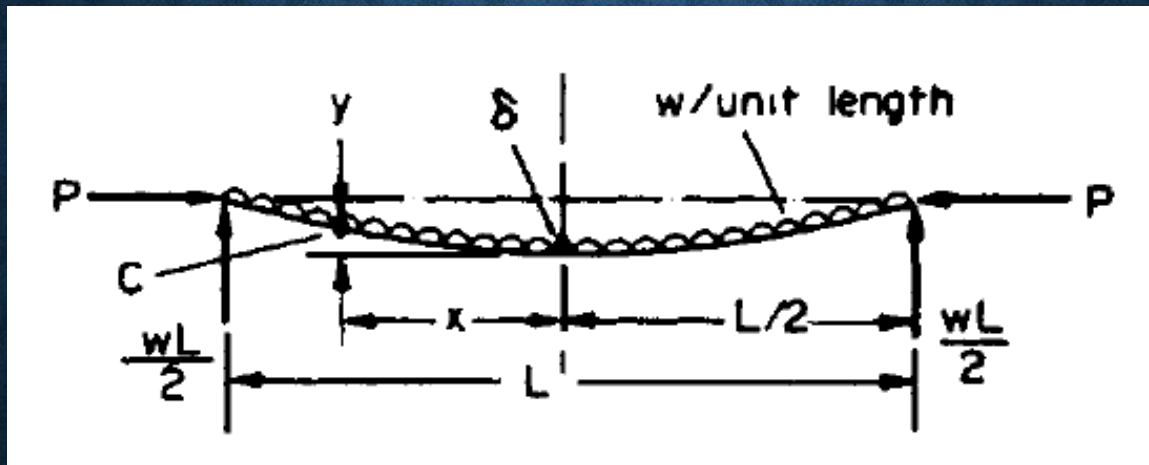
$$y_{\max} = \frac{W}{2nP} \left[\tan \frac{nL}{2} - \frac{nL}{2} \right] \dots \dots \dots \text{Eq.28}$$

- The maximum B.M acting on the strut is at the same position and is given by

$$\begin{aligned} M_{\max} &= -Py_{\max} - \frac{W}{2} \frac{L}{2} \\ &= -\frac{W}{2n} \tan \frac{nL}{2} \dots \dots \dots \text{Eq.29} \end{aligned}$$

b. Uniformly distributed load

Consider now the uniformly loaded strut of **Figure 2.12** with the origin again selected at the centre but y measured from the maximum deflected position.



Thus the maximum deflection δ , when $y = 0$ and $x = 0$, is given by

$$\delta = y_{\max} = \frac{w}{n^2 p} \left[\left(\sec \frac{nL}{2} - 1 \right) - \frac{n^2 L^2}{8} \right]$$

and the maximum B.M. is

$$M_{\max} = P\delta + \frac{wL^2}{8} = \frac{w}{n^2} \left(\sec \frac{nL}{2} - 1 \right)$$

- The maximum moment and also can be given by :

$$M = A \cosh nx + B \sinh nx + \text{etc.}$$

$$M_{\max} = \frac{w}{n^2} \left(\operatorname{sech} \frac{nL}{2} - 1 \right)$$

- **Refer** Mechanics of Materials, E.J. Hearn. Page 45-48 for the derivation of the above equations.